

ANALYSTLAB AFRICA

Machine Learning Internship Programme

Dataset Inspection Report

1. Dataset Overview

Attribute	Value
Source	Kaggle — Telco Customer Churn (blastchar/telco-customer-churn)
Rows (customers)	7,043
Columns (fields)	21
Unit of observation	One row per customer, identified by customerID
Target variable	Churn (Yes / No)

2. Data Types

The 21 columns break down as follows:

Type	Columns
Identifier	customerID (string, unique, no predictive value)
Numeric	SeniorCitizen (0/1 flag), tenure (integer, months), MonthlyCharges (float), TotalCharges (stored as text — see Section 3)
Categorical (binary)	gender, Partner, Dependents, PhoneService, PaperlessBilling, Churn
Categorical (multi-class)	MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies, Contract, PaymentMethod

3. Missing Values

A direct null check (`df.isnull().sum()`) initially reports zero missing values across all 21 columns. This is misleading: TotalCharges is loaded as a text column, and 11 rows contain a blank string rather than a number. Converting TotalCharges to numeric with coercion reveals 11 missing values.

Investigating these 11 rows shows every one of them has tenure = 0 — these are brand-new customers who have not yet completed a full billing cycle, so no total charge has accrued yet. This is a structural, explainable gap rather than random missing data.

Recommendation: impute these 11 TotalCharges values as 0 (or equivalently $\text{MonthlyCharges} \times \text{tenure}$), rather than dropping the rows, since they still carry a valid Churn label and represent a real (and arguably high-risk, since new) customer segment.

4. Duplicates

- Fully duplicated rows: 0
- Duplicate customerID values: 0

No duplicate-record cleanup is required.

5. Target Variable Distribution (Class Balance)

Class	Count	Percentage
No (did not churn)	5,174	73.5%
Yes (churned)	1,869	26.5%

The dataset is moderately imbalanced (roughly 3:1). This does not block modelling, but it does mean plain accuracy is a poor evaluation metric, and techniques such as class weighting, stratified sampling, or resampling (e.g. SMOTE) should be considered during model development.

6. Categorical Field Value Checks

Spot checks on key categorical fields confirm clean, consistent category labels with no stray whitespace, casing inconsistencies, or unexpected values:

- gender: Female, Male
- Contract: Month-to-month, One year, Two year
- InternetService: DSL, Fiber optic, No
- PaymentMethod: Electronic check, Mailed check, Bank transfer (automatic), Credit card (automatic)

7. Summary of Required Preprocessing

- Convert TotalCharges from text to numeric; impute the 11 resulting NaNs as 0.
- Drop customerID before modelling (identifier, not a feature).
- One-hot / dummy-encode multi-class categorical fields; binary-encode Yes/No and gender fields.
- Scale numeric fields (tenure, MonthlyCharges, TotalCharges) for algorithms sensitive to feature scale.
- Use a stratified train/test split to preserve the ~73.5% / 26.5% class ratio in both sets.
- Address class imbalance explicitly at the modelling stage (class weights and/or resampling).